

WORKED EXAMPLE

Learn a threshold neuron from mistakes

Lecture 3, PDF page 41: labels are +1 and -1, with a unit-size update.

Original study data, always visited in this order: A = (1,0), target +1; B = (0,1), target -1; C = (1,1), target +1.

$$X = (1, x_1, x_2), \quad W = (b, w_1, w_2), \quad s = W^T X = b + w_1 x_1 + w_2 x_2$$

The leading 1 is a bookkeeping input for the bias. It is not another measured feature.

$$\hat{y} = +1 \text{ if } s \geq 0; \quad \hat{y} = -1 \text{ if } s < 0$$

The slide does not define $\text{sign}(0)$. We choose +1 explicitly. These labels differ from the 0/1 labels used in logistic regression.

WORKED EXAMPLE

Update only after a mistaken prediction

An epoch is one complete pass through A, B, C in that order.

$$\hat{y} \neq y: \quad W_{\text{new}} = W_{\text{old}} + yX$$

$$\hat{y} = y: \quad W_{\text{new}} = W_{\text{old}}$$

$$W_{\text{start}} = (0, 0, 0)$$

Use the newest weights for the very next example. The update is not a batch average and does not wait until the epoch ends.

Stop only after an entire epoch has no classification errors. With our tie rule, score zero is correct for target +1 and incorrect for target -1.

WORKED EXAMPLE

Epoch 1, point A: keep the weights

Visit 1 of 12. Target label +1; prediction follows the stated sign rule.

$$W_{\text{old}} = (0, 0, 0), \quad X = (1, 1, 0), \quad y = +1$$

$$s = (0) + (0)(1) + (0)(0) = 0$$

$$\hat{y} = +1, \quad y = +1 \Rightarrow \text{correct}$$

$$W_{\text{new}} = W_{\text{old}} = (0, 0, 0)$$

No update: a correctly classified point does not change any parameter.

WORKED EXAMPLE

Epoch 1, point B: correct the mistake

Visit 2 of 12. Target label -1; prediction follows the stated sign rule.

$$W_{\text{old}} = (0, 0, 0), \quad X = (1, 0, 1), \quad y = -1$$

$$s = (0) + (0)(0) + (0)(1) = 0$$

$$\hat{y} = +1, \quad y = -1 \Rightarrow \text{mistake}$$

$$yX = (-1)(1, 0, 1) = (-1, 0, -1)$$

$$W_{\text{new}} = (0, 0, 0) + (-1, 0, -1) = (-1, 0, -1)$$

Carry these updated values into the next visit. The bias changes too because the first input component is 1.

WORKED EXAMPLE

Epoch 1, point C: correct the mistake

Visit 3 of 12. Target label +1; prediction follows the stated sign rule.

$$W_{\text{old}} = (-1, 0, -1), \quad X = (1, 1, 1), \quad y = +1$$

$$s = (-1) + (0)(1) + (-1)(1) = -2$$

$$\hat{y} = -1, \quad y = +1 \Rightarrow \text{mistake}$$

$$yX = (1)(1, 1, 1) = (1, 1, 1)$$

$$W_{\text{new}} = (-1, 0, -1) + (1, 1, 1) = (0, 1, 0)$$

Carry these updated values into the next visit. The bias changes too because the first input component is 1.

WORKED EXAMPLE

Epoch 2, point A: keep the weights

Visit 4 of 12. Target label +1; prediction follows the stated sign rule.

$$W_{\text{old}} = (0, 1, 0), \quad X = (1, 1, 0), \quad y = +1$$

$$s = (0) + (1)(1) + (0)(0) = 1$$

$$\hat{y} = +1, \quad y = +1 \Rightarrow \text{correct}$$

$$W_{\text{new}} = W_{\text{old}} = (0, 1, 0)$$

No update: a correctly classified point does not change any parameter.

WORKED EXAMPLE

Epoch 2, point B: correct the mistake

Visit 5 of 12. Target label -1; prediction follows the stated sign rule.

$$W_{\text{old}} = (0, 1, 0), \quad X = (1, 0, 1), \quad y = -1$$

$$s = (0) + (1)(0) + (0)(1) = 0$$

$$\hat{y} = +1, \quad y = -1 \Rightarrow \text{mistake}$$

$$yX = (-1)(1, 0, 1) = (-1, 0, -1)$$

$$W_{\text{new}} = (0, 1, 0) + (-1, 0, -1) = (-1, 1, -1)$$

Carry these updated values into the next visit. The bias changes too because the first input component is 1.

WORKED EXAMPLE

Epoch 2, point C: correct the mistake

Visit 6 of 12. Target label +1; prediction follows the stated sign rule.

$$W_{\text{old}} = (-1, 1, -1), \quad X = (1, 1, 1), \quad y = +1$$

$$s = (-1) + (1)(1) + (-1)(1) = -1$$

$$\hat{y} = -1, \quad y = +1 \Rightarrow \text{mistake}$$

$$yX = (1)(1, 1, 1) = (1, 1, 1)$$

$$W_{\text{new}} = (-1, 1, -1) + (1, 1, 1) = (0, 2, 0)$$

Carry these updated values into the next visit. The bias changes too because the first input component is 1.

WORKED EXAMPLE

Epoch 3, point A: keep the weights

Visit 7 of 12. Target label +1; prediction follows the stated sign rule.

$$W_{\text{old}} = (0, 2, 0), \quad X = (1, 1, 0), \quad y = +1$$

$$s = (0) + (2)(1) + (0)(0) = 2$$

$$\hat{y} = +1, \quad y = +1 \Rightarrow \text{correct}$$

$$W_{\text{new}} = W_{\text{old}} = (0, 2, 0)$$

No update: a correctly classified point does not change any parameter.

WORKED EXAMPLE

Epoch 3, point B: correct the mistake

Visit 8 of 12. Target label -1; prediction follows the stated sign rule.

$$W_{\text{old}} = (0, 2, 0), \quad X = (1, 0, 1), \quad y = -1$$

$$s = (0) + (2)(0) + (0)(1) = 0$$

$$\hat{y} = +1, \quad y = -1 \Rightarrow \text{mistake}$$

$$yX = (-1)(1, 0, 1) = (-1, 0, -1)$$

$$W_{\text{new}} = (0, 2, 0) + (-1, 0, -1) = (-1, 2, -1)$$

Carry these updated values into the next visit. The bias changes too because the first input component is 1.

WORKED EXAMPLE

Epoch 3, point C: keep the weights

Visit 9 of 12. Target label +1; prediction follows the stated sign rule.

$$W_{\text{old}} = (-1, 2, -1), \quad X = (1, 1, 1), \quad y = +1$$

$$s = (-1) + (2)(1) + (-1)(1) = 0$$

$$\hat{y} = +1, \quad y = +1 \Rightarrow \text{correct}$$

$$W_{\text{new}} = W_{\text{old}} = (-1, 2, -1)$$

No update: a correctly classified point does not change any parameter.

WORKED EXAMPLE

Epoch 4, point A: keep the weights

Visit 10 of 12. Target label +1; prediction follows the stated sign rule.

$$W_{\text{old}} = (-1, 2, -1), \quad X = (1, 1, 0), \quad y = +1$$

$$s = (-1) + (2)(1) + (-1)(0) = 1$$

$$\hat{y} = +1, \quad y = +1 \Rightarrow \text{correct}$$

$$W_{\text{new}} = W_{\text{old}} = (-1, 2, -1)$$

No update: a correctly classified point does not change any parameter.

WORKED EXAMPLE

Epoch 4, point B: keep the weights

Visit 11 of 12. Target label -1; prediction follows the stated sign rule.

$$W_{\text{old}} = (-1, 2, -1), \quad X = (1, 0, 1), \quad y = -1$$

$$s = (-1) + (2)(0) + (-1)(1) = -2$$

$$\hat{y} = -1, \quad y = -1 \Rightarrow \text{correct}$$

$$W_{\text{new}} = W_{\text{old}} = (-1, 2, -1)$$

No update: a correctly classified point does not change any parameter.

WORKED EXAMPLE

Epoch 4, point C: keep the weights

Visit 12 of 12. Target label +1; prediction follows the stated sign rule.

$$W_{\text{old}} = (-1, 2, -1), \quad X = (1, 1, 1), \quad y = +1$$

$$s = (-1) + (2)(1) + (-1)(1) = 0$$

$$\hat{y} = +1, \quad y = +1 \Rightarrow \text{correct}$$

$$W_{\text{new}} = W_{\text{old}} = (-1, 2, -1)$$

No update: a correctly classified point does not change any parameter.

WORKED EXAMPLE

Why the mistake update helps this point

Examine the first mistaken visit: B in epoch 1.

$$X_B = (1, 0, 1), \quad y_B = -1, \quad W_{\text{old}} = (0, 0, 0)$$

$$W_{\text{new}} = (-1, 0, -1), \quad s_{\text{new}} = -1 + 0(0) - 1(1) = -2$$

$$\Delta s = (yX)^T X = y\|X\|^2 = (-1)(1^2 + 0^2 + 1^2) = -2$$

Squared length means the sum of squared components. The score moves downward toward the negative target. A positive target moves its own score upward. Other points can become wrong again, which is why we repeat epochs.

Do not replace the mistake test by $y \text{ times score} \leq 0$ under this tie convention: a zero score with target +1 is already correct.

WORKED EXAMPLE

Verify the final neuron and stop

The fourth epoch has zero errors; the third still had one.

epoch	mistakes	weights at end
1	2	(0,1,0)
2	2	(0,2,0)
3	1	(-1,2,-1)
4	0	(-1,2,-1)

$$s = -1 + 2x_1 - x_2$$

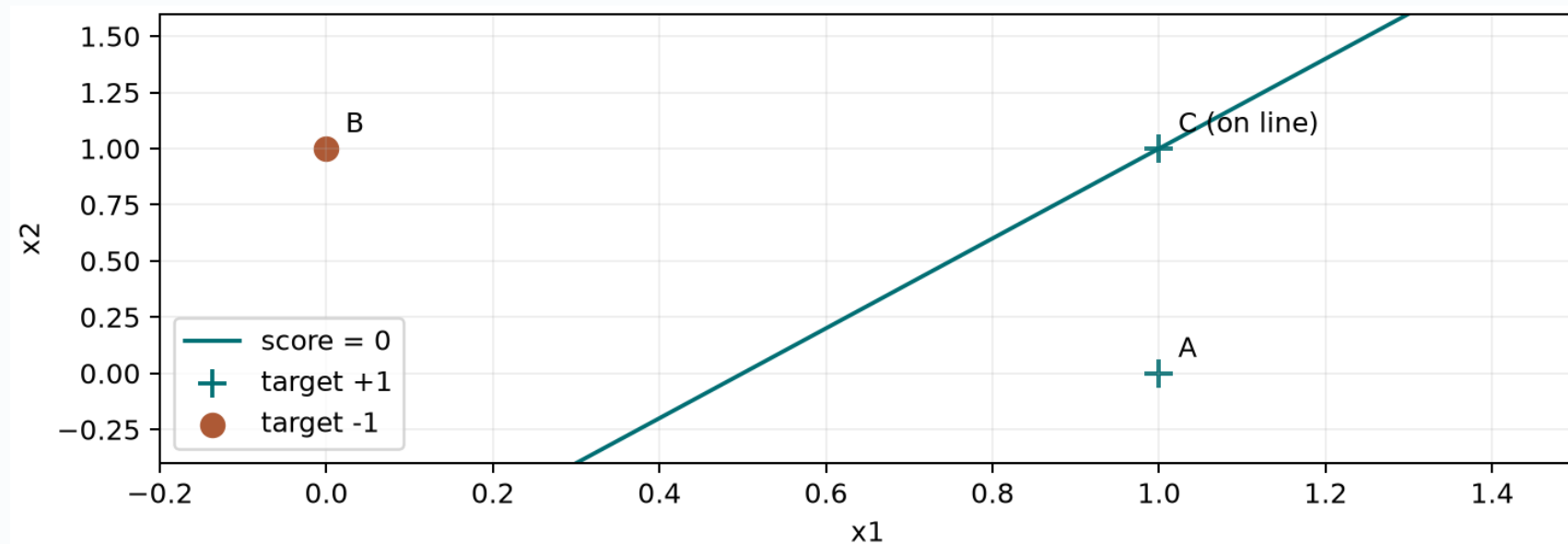
$$A : s = 1 \Rightarrow +1; \quad B : s = -2 \Rightarrow -1; \quad C : s = 0 \Rightarrow +1$$

Five updates occurred over twelve visits. A visit, a mistake update and an epoch are different counts.

WORKED EXAMPLE

Picture the final separating boundary

Points on the line receive +1 because $\text{sign}(0) = +1$.



The boundary is $2x_1 - x_2 - 1 = 0$. C lies exactly on it; our tie convention matters.

Lecture 3, pages 53-54: a single perceptron cannot solve nonseparable data such as XOR. Repeating the loop then need not produce an error-free epoch.

INDEPENDENT PRACTICE

Your turn: two points, one measured feature

Keep +1/-1 labels, $\text{sign}(0) = +1$ and the unit mistake update.

$$X = (1, x), \quad W = (b, w), \quad W_{\text{start}} = (0, -1)$$

Fixed order: first $x = -1$ with target -1 ; then $x = 2$ with target $+1$.

Show the old weights, score, predicted label and resulting weights for every visit. Repeat until a complete epoch has no errors.

How many visits, mistake updates and epochs are required? Explain why the first epoch alone is not enough.

WORKED ANSWER

Practice answer: correct both first-epoch errors

The second point uses the weights already changed by the first.

$$x = -1: \quad s = 0 + (-1)(-1) = 1 \Rightarrow \hat{y} = +1 \neq -1$$

$$W = (0, -1) + (-1)(1, -1) = (0, -1) + (-1, 1) = (-1, 0)$$

$$x = 2: \quad s = -1 + 0(2) = -1 \Rightarrow \hat{y} = -1 \neq +1$$

$$W = (-1, 0) + (+1)(1, 2) = (-1, 0) + (1, 2) = (0, 2)$$

Both visits made an update. Even though the final weights may now be correct, the stopping rule requires another complete pass.

WORKED ANSWER

Practice answer: check the whole second epoch

The first correct point is not yet an error-free epoch.

$$x = -1 : \quad s = 0 + 2(-1) = -2 \Rightarrow \hat{y} = -1 = y$$

W stays $(0, 2)$

$$x = 2 : \quad s = 0 + 2(2) = 4 \Rightarrow \hat{y} = +1 = y$$

W stays $(0, 2)$

Now stop: 2 epochs, 4 visits, 2 mistake updates. Remember to update the bias, preserve visit order, and distinguish a score from its predicted label.